

User to Operator: How to Run a Tor Relay and Defend Online Freedom

Dan Salmon



<https://danthesalmon.com/files/2026-tor-talk.pdf>

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Intro

- Pentester by trade
- Collector of side projects
- **Relay** operator ~3 years
- Not a member of the Tor Project

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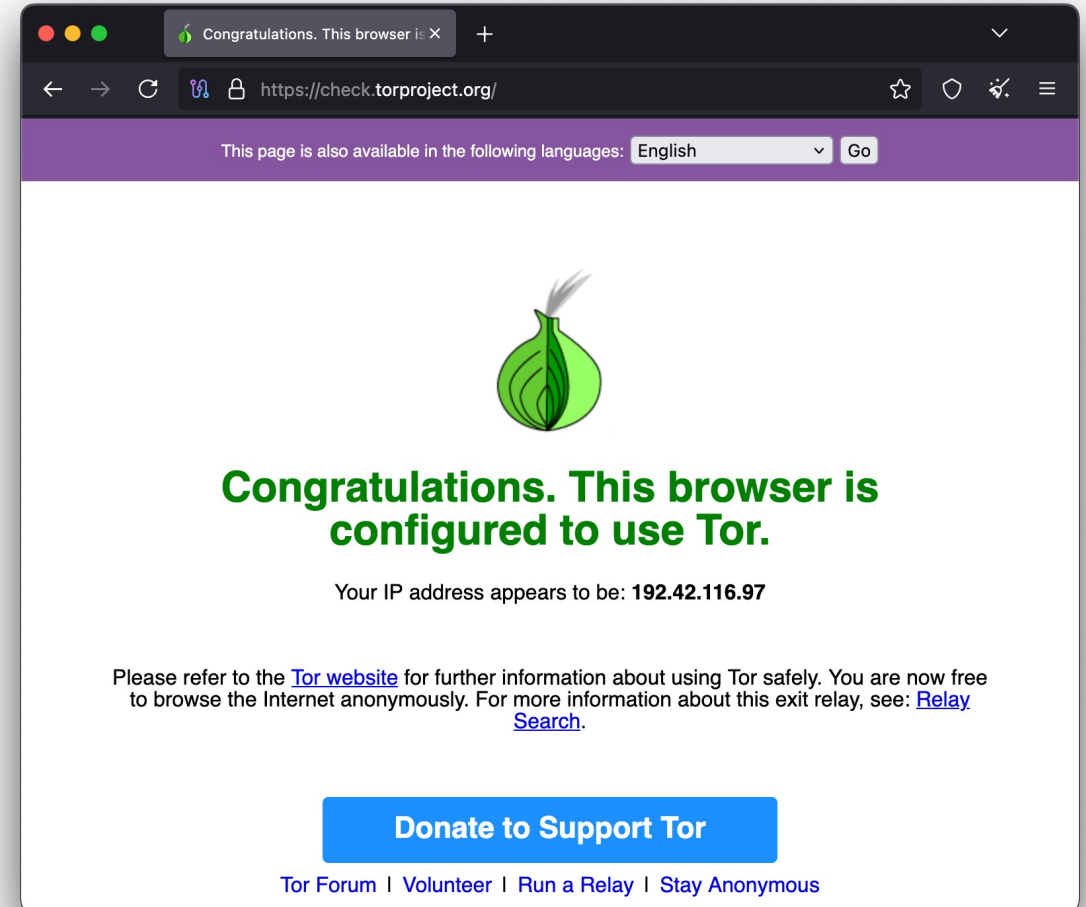
 sa7mon

 danthesalmon.com

What is Tor

What is Tor

- Tor is free and open-source software (**FOSS**) providing anonymous internet routing
- Network of volunteer-run servers (**relays**)
- Users connect via applications like Tor Browser

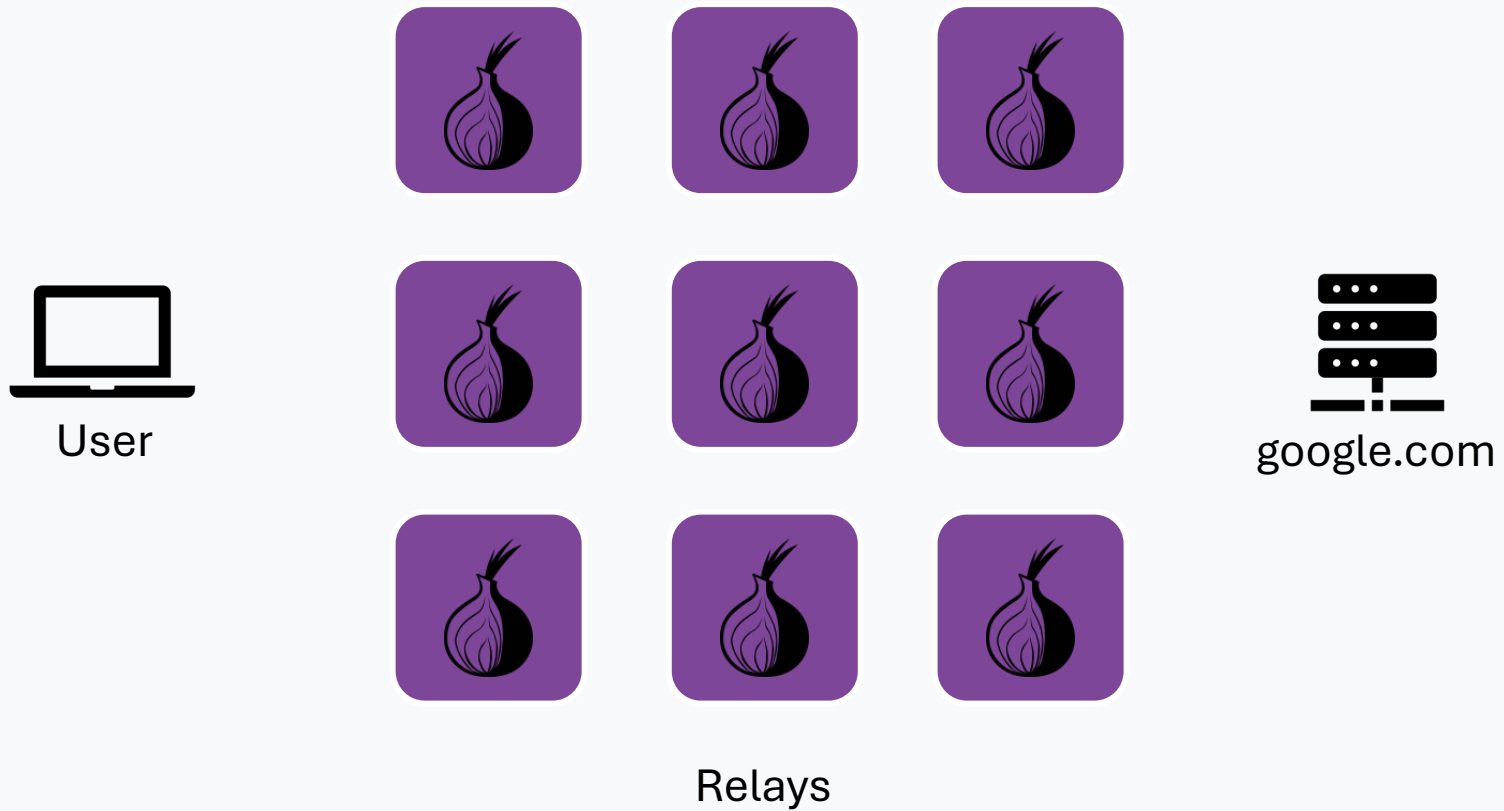


Tor Browser

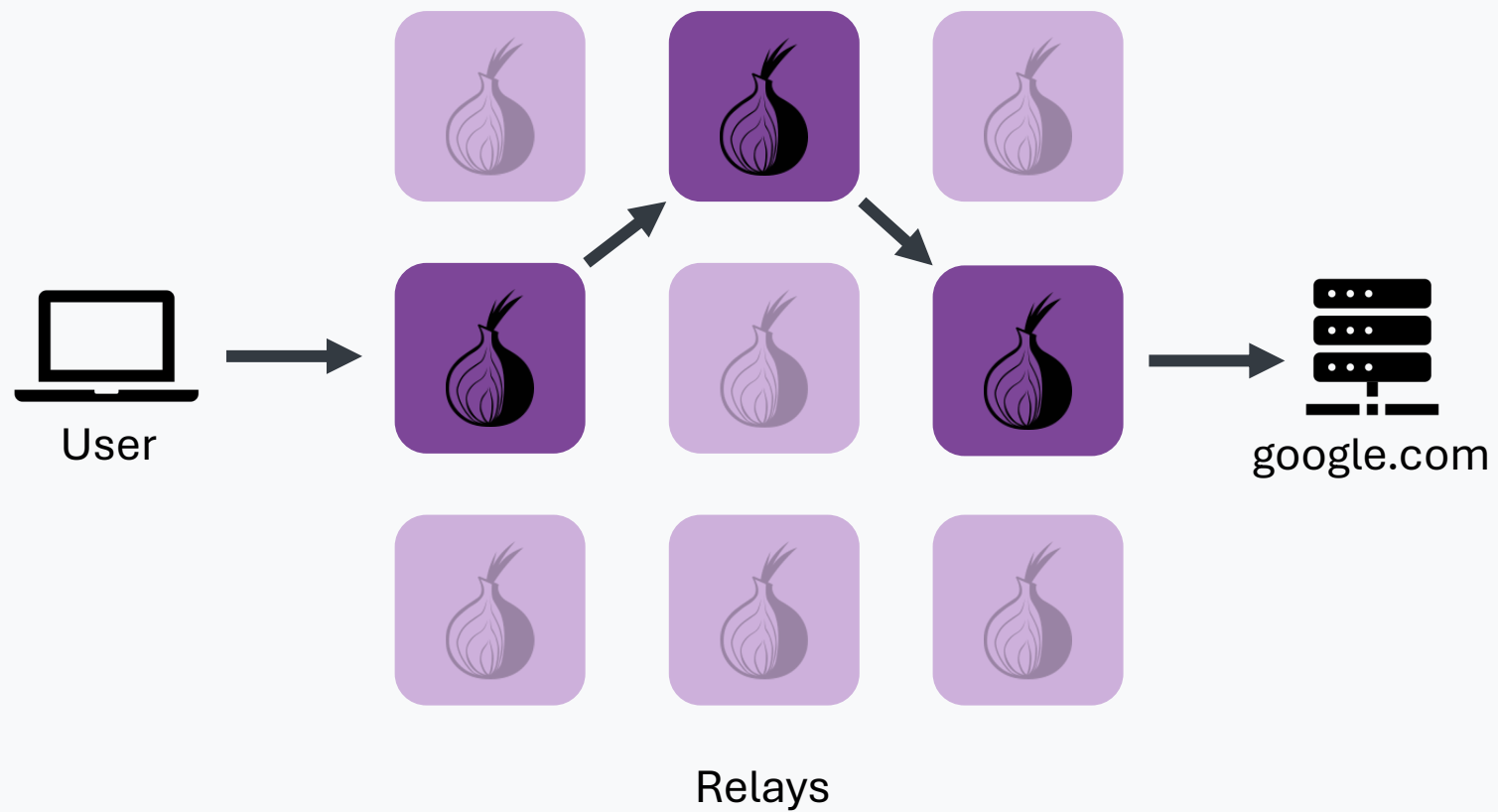
What is Tor

- When using the Internet, traffic can be **identified** by adversaries
 - Internet Service Provider (ISPs)
 - Intelligence agencies
- Traffic identification enables **surveillance** and **censorship**
- Tor routes traffic through 3 relays, encrypting the data 3x
- Each relay decrypts a layer, like peeling an onion ⇒ "**onion routing**"

Tor Circuit

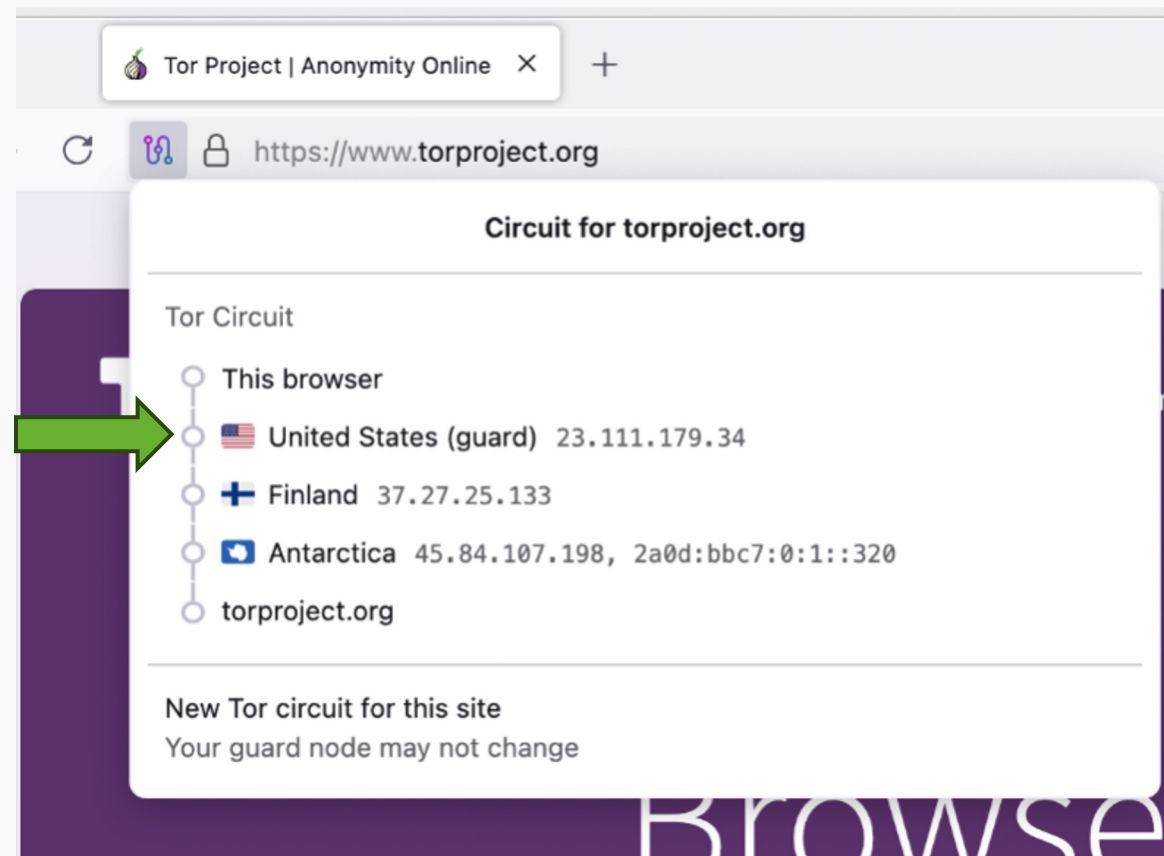


Tor Circuit



Relay Type: Guard

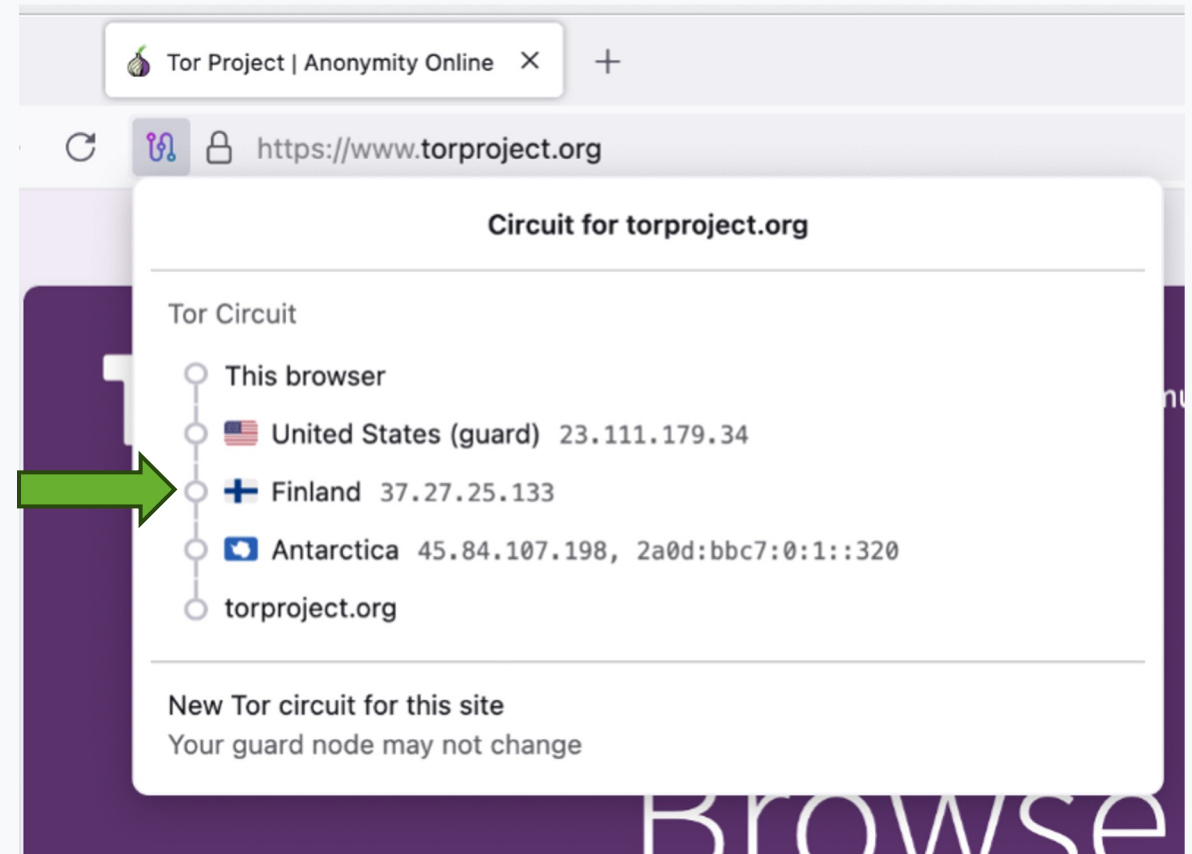
- 1st hop
- Fast & stable
- Can see the user's IP and middle relay IP



Tor Browser

Relay Type: Middle

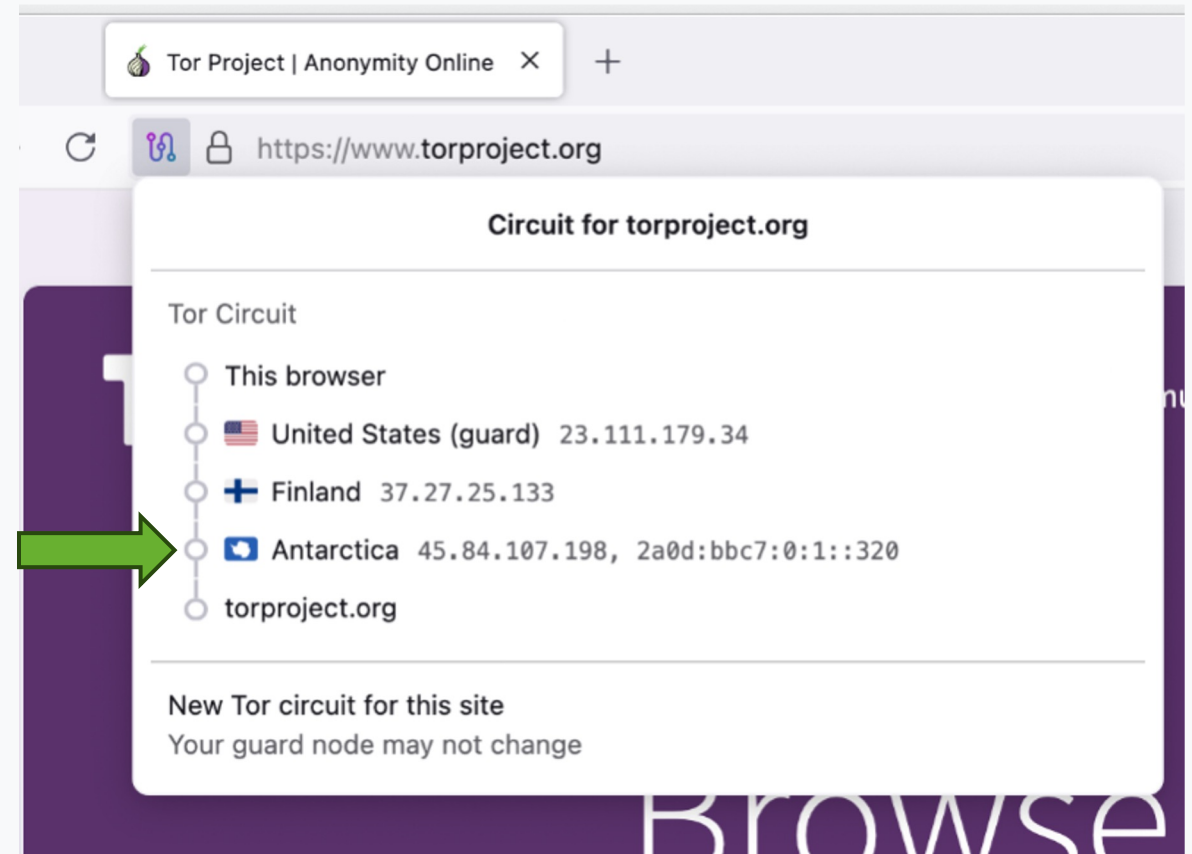
- 2nd hop
- Can only see the guard and exit relay IPs
- Least likely to receive legal complaints
- Focus of this talk



Tor Browser

Relay Type: Exit

- 3rd hop
- Can only see the middle relay and destination IPs
- Most likely to receive legal complaints



Tor Browser

Relay Type: Bridge

- Replaces guard relay as 1st hop
- Used when Tor is blocked in user's region
- Hidden: IP address **not** listed in public directory
- Pluggable Transports add obfuscation

Pluggable Transport	Looks Like
obfs4	Random data
meek	Connection to a big cloud provider
snowflake	Video call
WebTunnel	Encrypted web traffic (HTTPS)

Should You Run a Relay

Legal Considerations*

- IP will get added to block lists
- Contact email is public
 - Be aware of persona mixing
- Not immune to attacks



Defending the Tor network: Mitigating IP spoofing against Tor

by gus | November 8, 2024

<https://blog.torproject.org/defending-tor-mitigating-IP-spoofing>

Details for: pushypebble ●

Configuration

Nickname ●

pushypebble

OR Addresses ●

172.81.131.87:443

[2605:aa80:c007:1736::]:443

Contact

salmon at pm dot me

Dir Address

none

Exit Addresses

none

Advertised Bandwidth

6.57 MiB/s

* not a lawyer

Why You Should Run a Relay

- More relays make the network:
 - faster (therefore more usable)
 - more stable in case of outages
 - more robust against attacks

“The Tor network's anonymity comes in part from diversity, so if you can run your own relay, you will be improving Tor's anonymity more than by donating”

- <https://support.torproject.org/faq/>

Why You Should Run a Relay

- It's fun!
- Also, t-shirt



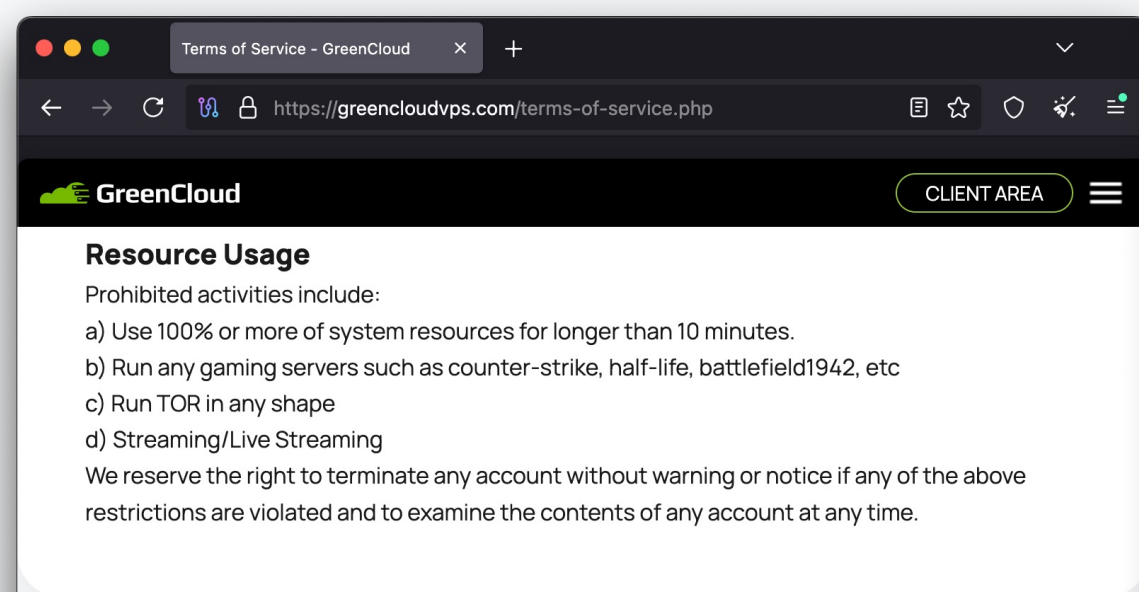
Choosing a Hosting Provider

Choosing a Provider

- Looking for Virtual Private Server (VPS) hosting
 - aka "cloud" providers
- Colocation (colo) also an option

Choosing a Provider: Terms of Service

No Tor usage allowed



The screenshot shows a web browser window with the title "Terms of Service - GreenCloud". The address bar displays "https://greencloudvps.com/terms-of-service.php". The page header includes the "GreenCloud" logo and a "CLIENT AREA" button. The main content area is titled "Resource Usage" and lists prohibited activities. It states: "Prohibited activities include: a) Use 100% or more of system resources for longer than 10 minutes. b) Run any gaming servers such as counter-strike, half-life, battlefield1942, etc c) Run TOR in any shape d) Streaming/Live Streaming". It also includes a disclaimer: "We reserve the right to terminate any account without warning or notice if any of the above restrictions are violated and to examine the contents of any account at any time."

Terms of Service - GreenCloud

https://greencloudvps.com/terms-of-service.php

GreenCloud

CLIENT AREA

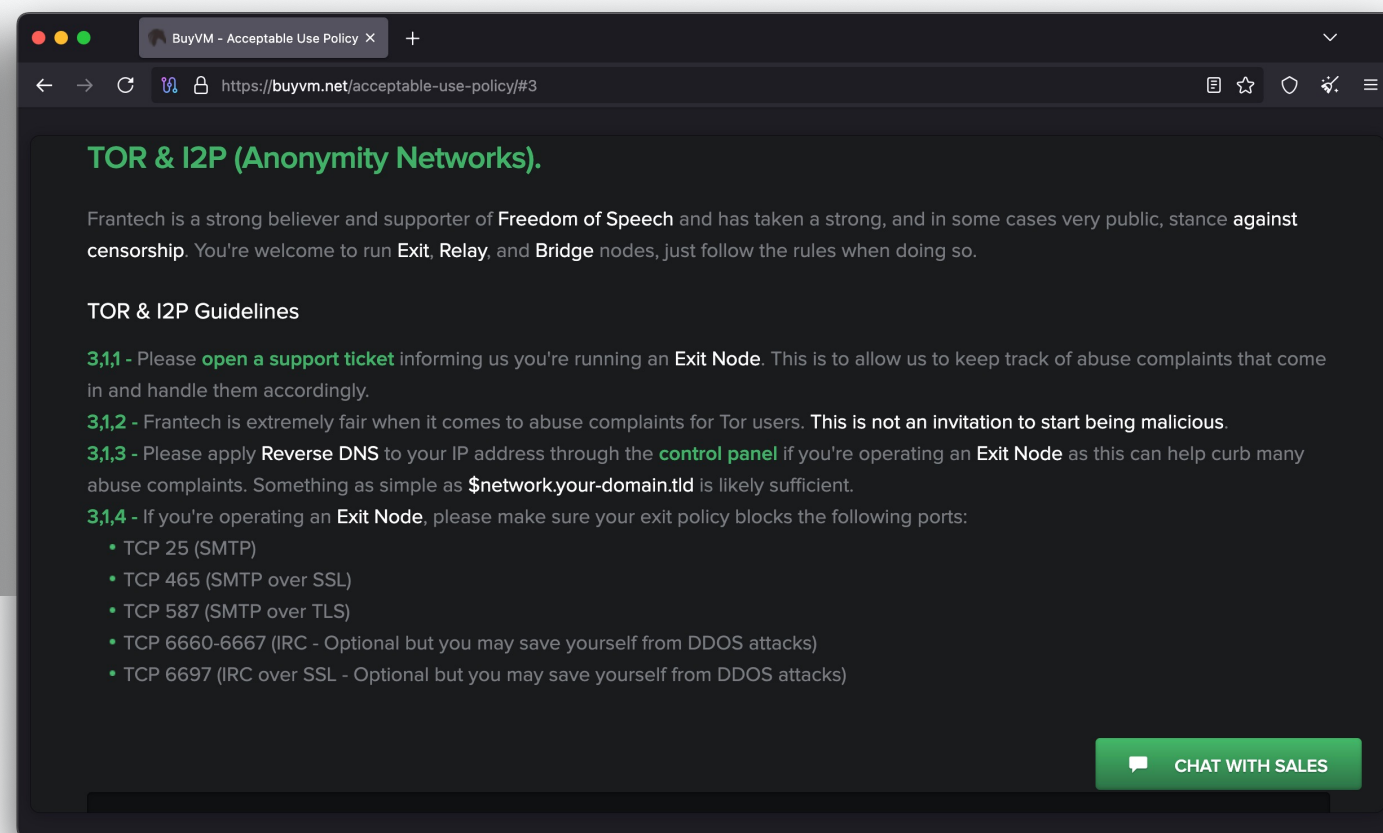
Resource Usage

Prohibited activities include:

- a) Use 100% or more of system resources for longer than 10 minutes.
- b) Run any gaming servers such as counter-strike, half-life, battlefield1942, etc
- c) Run TOR in any shape
- d) Streaming/Live Streaming

We reserve the right to terminate any account without warning or notice if any of the above restrictions are violated and to examine the contents of any account at any time.

All relays allowed, notify if running Exit



The screenshot shows a web browser window with the title "BuyVM - Acceptable Use Policy". The address bar displays "https://buyvm.net/acceptable-use-policy/#3". The page content is titled "TOR & I2P (Anonymity Networks)." and includes a paragraph about Frantech's stance on freedom of speech and censorship. It then lists "TOR & I2P Guidelines" with four numbered items (3,1,1 to 3,1,4) detailing requirements for running Exit Nodes, including opening support tickets, applying Reverse DNS, and blocking specific ports. A "CHAT WITH SALES" button is visible in the bottom right corner.

BuyVM - Acceptable Use Policy

https://buyvm.net/acceptable-use-policy/#3

TOR & I2P (Anonymity Networks).

Frantech is a strong believer and supporter of **Freedom of Speech** and has taken a strong, and in some cases very public, stance **against** censorship. You're welcome to run **Exit**, **Relay**, and **Bridge** nodes, just follow the rules when doing so.

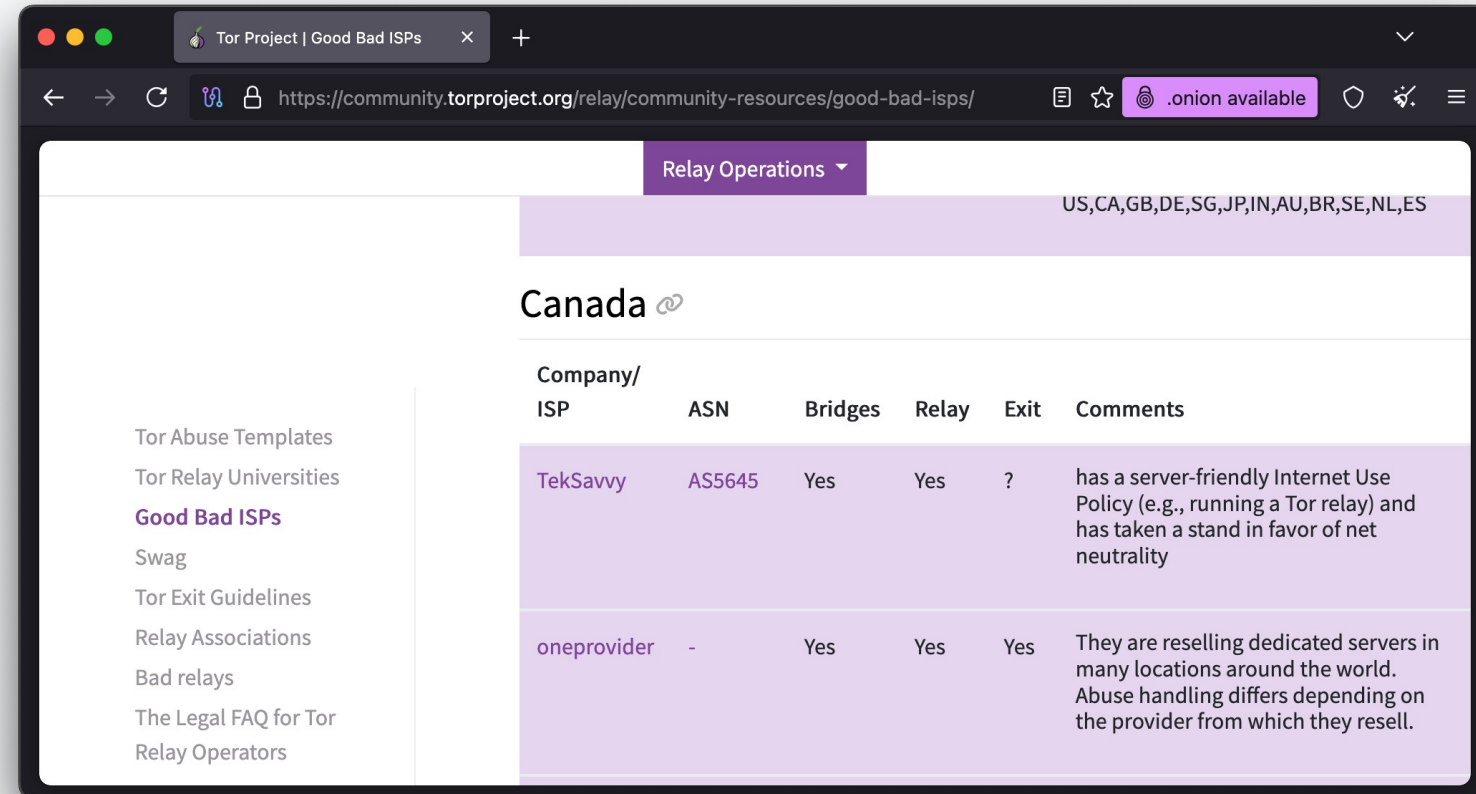
TOR & I2P Guidelines

- 3,1,1** - Please **open a support ticket** informing us you're running an **Exit Node**. This is to allow us to keep track of abuse complaints that come in and handle them accordingly.
- 3,1,2** - Frantech is extremely fair when it comes to abuse complaints for Tor users. **This is not an invitation to start being malicious.**
- 3,1,3** - Please apply **Reverse DNS** to your IP address through the **control panel** if you're operating an **Exit Node** as this can help curb many abuse complaints. Something as simple as **\$network.your-domain.tld** is likely sufficient.
- 3,1,4** - If you're operating an **Exit Node**, please make sure your exit policy blocks the following ports:
 - TCP 25 (SMTP)
 - TCP 465 (SMTP over SSL)
 - TCP 587 (SMTP over TLS)
 - TCP 6660-6667 (IRC - Optional but you may save yourself from DDOS attacks)
 - TCP 6697 (IRC over SSL - Optional but you may save yourself from DDOS attacks)

CHAT WITH SALES

Choosing a Provider: Terms of Service

- Check & update **Good Bad ISPs** list



The screenshot shows a web browser window with the URL <https://community.torproject.org/relay/community-resources/good-bad-isps/>. The page has a dark theme. On the left is a sidebar with links: Tor Abuse Templates, Tor Relay Universities, **Good Bad ISPs** (highlighted), Swag, Tor Exit Guidelines, Relay Associations, Bad relays, The Legal FAQ for Tor, and Relay Operators. The main content area has a 'Relay Operations' dropdown menu and a list of countries: US, CA, GB, DE, SG, JP, IN, AU, BR, SE, NL, ES. 'Canada' is selected and highlighted. Below it is a table of ISPs in Canada.

Company/ ISP	ASN	Bridges	Relay	Exit	Comments
TekSavvy	AS5645	Yes	Yes	?	has a server-friendly Internet Use Policy (e.g., running a Tor relay) and has taken a stand in favor of net neutrality
oneprovider	-	Yes	Yes	Yes	They are reselling dedicated servers in many locations around the world. Abuse handling differs depending on the provider from which they resell.

<https://community.torproject.org/relay/community-resources/good-bad-isps/>

Choosing a Provider: Network Diversity

- What is an ASN
- Large concentration of relays at cheap, popular providers
- A more diverse Tor network is more resilient to outages and attacks

Try to avoid the following hosters:

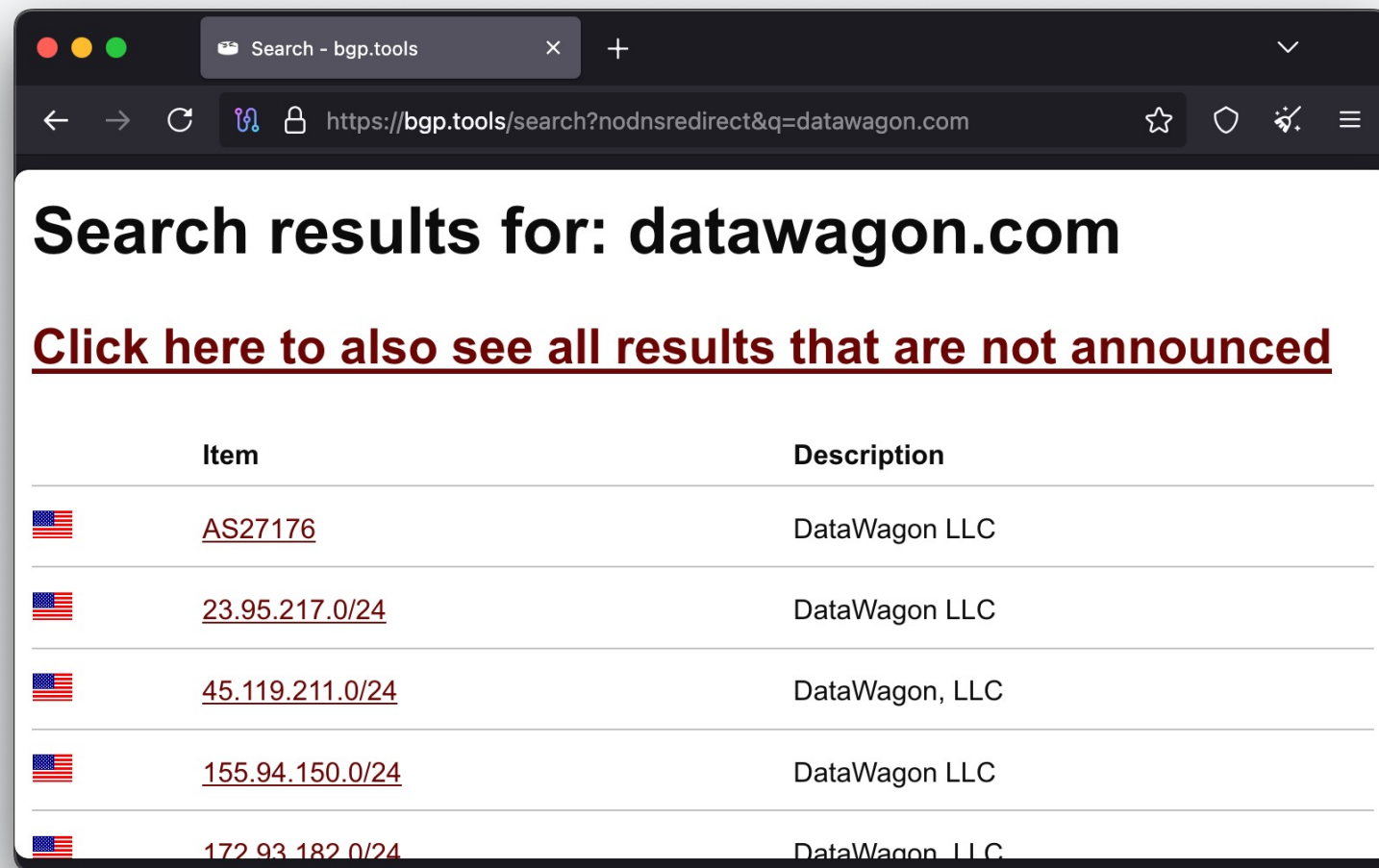
- Frantech / Ponynet / BuyVM (AS53667)
- OVH SAS / OVHcloud (AS16276)
- Online S.A.S. / Scaleway (AS12876)
- Hetzner Online GmbH (AS24940)
- IONOS SE (AS8560)
- netcup GmbH (AS197540)
- 1337 Services GmbH / RDP.sh (AS210558)

These hosts already have many Tor nodes being hosted there.

<https://community.torproject.org/relay/community-resources/good-bad-isps/>

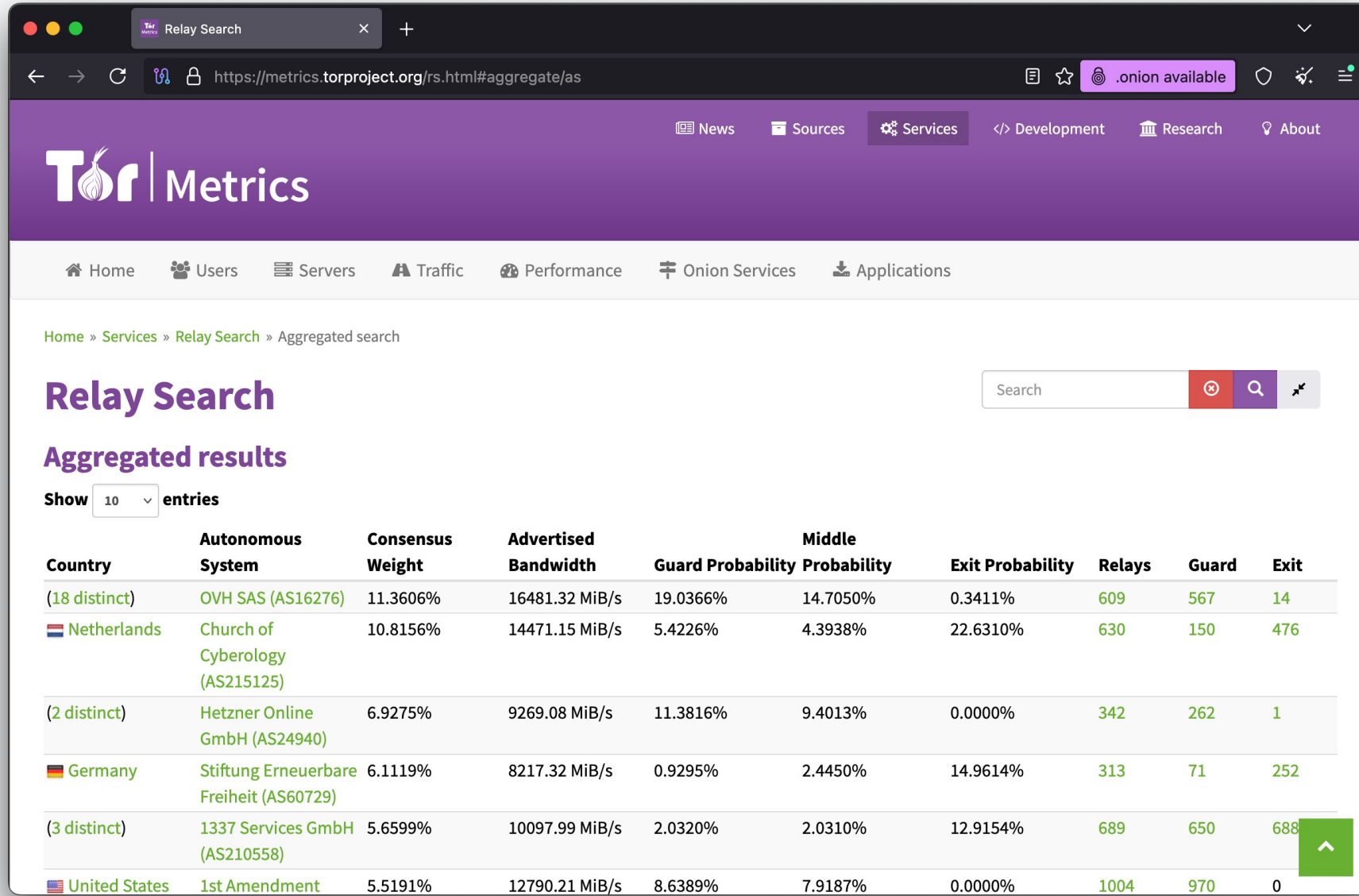
Choosing a Provider: Determining ASN

- Search help docs for “ASN” or “network”
- Search bgp.tools by name
- Email sales
- Just Deploy It
- Beware of resellers



	Item	Description
	<u>AS27176</u>	DataWagon LLC
	<u>23.95.217.0/24</u>	DataWagon LLC
	<u>45.119.211.0/24</u>	DataWagon, LLC
	<u>155.94.150.0/24</u>	DataWagon LLC
	<u>172.93.182.0/24</u>	DataWagon, LLC

Choosing a Provider: Network Diversity



The screenshot shows the Tor Metrics website's Relay Search interface. The page title is "Relay Search" and the subtitle is "Aggregated results". A search bar is visible on the right. Below the search bar, there is a table of results. The table has columns for Country, Autonomous System, Consensus Weight, Advertised Bandwidth, Guard Probability, Middle Probability, Exit Probability, Relays, Guard, and Exit. The results are sorted by consensus weight, with the highest being OVH SAS (AS16276) at 11.3606%.

Home » Services » Relay Search » Aggregated search

Relay Search

Search

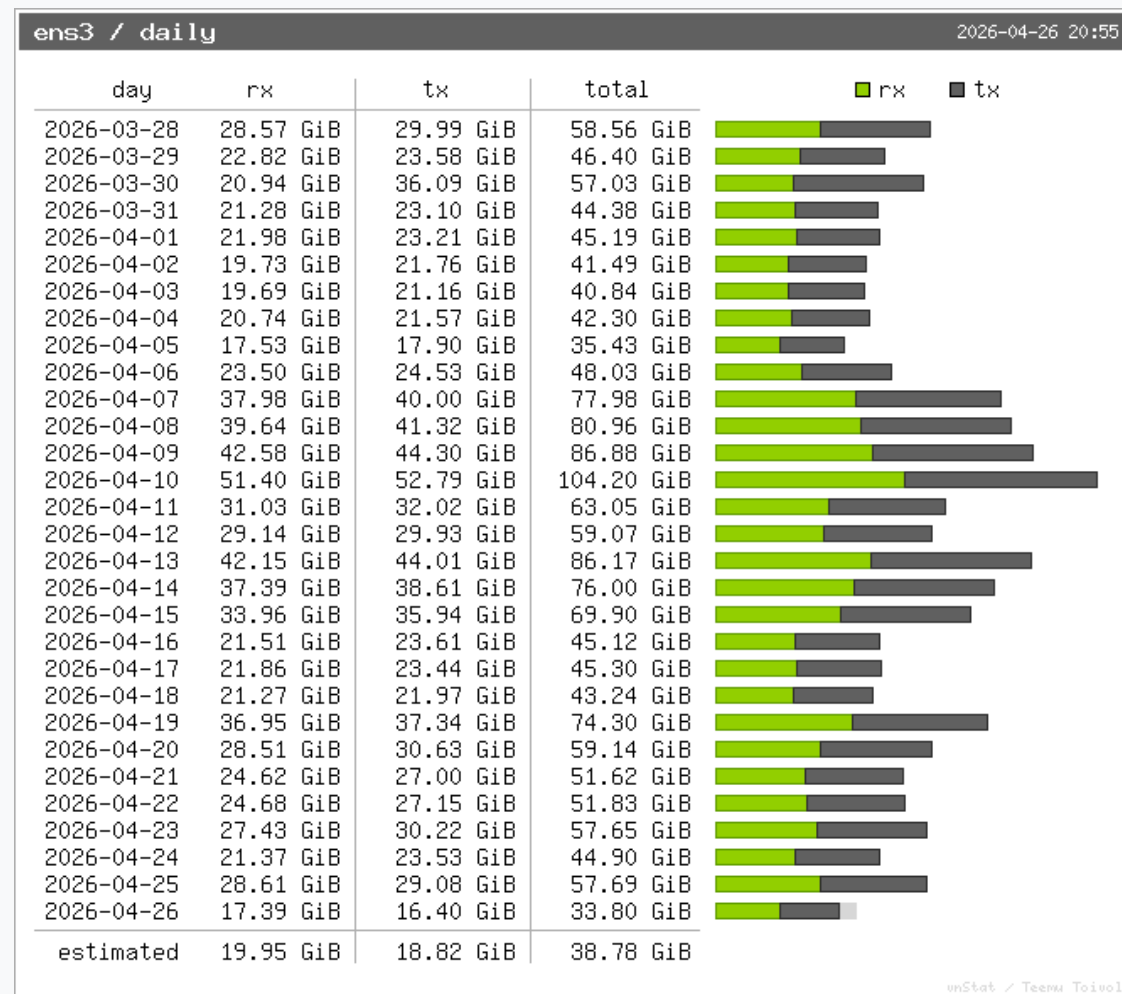
Aggregated results

Show 10 entries

Country	Autonomous System	Consensus Weight	Advertised Bandwidth	Guard Probability	Middle Probability	Exit Probability	Relays	Guard	Exit
(18 distinct)	OVH SAS (AS16276)	11.3606%	16481.32 MiB/s	19.0366%	14.7050%	0.3411%	609	567	14
 Netherlands	Church of Cyberology (AS215125)	10.8156%	14471.15 MiB/s	5.4226%	4.3938%	22.6310%	630	150	476
(2 distinct)	Hetzner Online GmbH (AS24940)	6.9275%	9269.08 MiB/s	11.3816%	9.4013%	0.0000%	342	262	1
 Germany	Stiftung Erneuerbare Freiheit (AS60729)	6.1119%	8217.32 MiB/s	0.9295%	2.4450%	14.9614%	313	71	252
(3 distinct)	1337 Services GmbH (AS210558)	5.6599%	10097.99 MiB/s	2.0320%	2.0310%	12.9154%	689	650	688
 United States	1st Amendment	5.5191%	12790.21 MiB/s	8.6389%	7.9187%	0.0000%	1004	970	0

Choosing a Provider: Traffic

- Amount of data transferred over period
- AKA “data transfer” or “bandwidth”
- Beware of “unlimited” traffic
- **Minimum:** 100GB/month
- **Recommended:** >2TB/month



Choosing an Operating System

- Diversity
- Limited by provider
- KVM access
 - ISO upload
 - Rescue Mode

OS Distribution (Relays)

Show 10 entries

Search:

OS	CW Fraction(%)	Exit(%)	Guard(%)	#Relays
Linux	90.5	88.5	91.5	9044
BSD	9.3	11.4	8.3	700
Windows	0	0	0	32
SunOS	0	0	0	8
Darwin	0	0	0	6
other	0	0	0	3

<https://nusenu.github.io/OrNetStats/#os-distribution-relays>

Unicorn

- Cheap server
- Unlimited traffic
- In uncommon ASN
- Can install any OS from .iso

Server Setup

Setup

- System hardening
- IPv6 for reachability and cool points
- `$package_manager install tor`
- Create `/etc/tor/torrc`

Nickname	myRelay
ContactInfo	e@mail.com
ORPort	443
ExitRelay	0
SocksPort	0

Running

What happens after `systemctl restart tor@default`

New Relay Lifecycle

- **Day 0-3**

The unmeasured phase

- **Day 3-8**

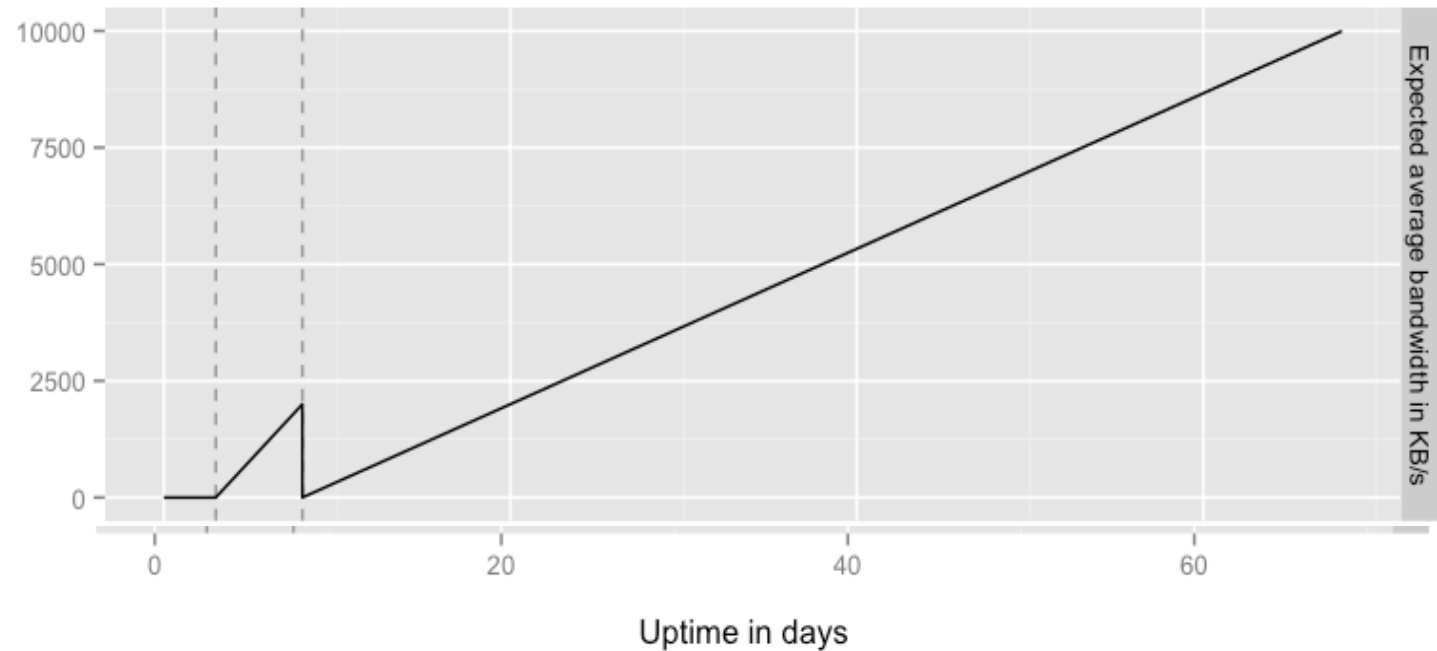
Remote measurement

- **Day 8-68**

Becoming guard relay

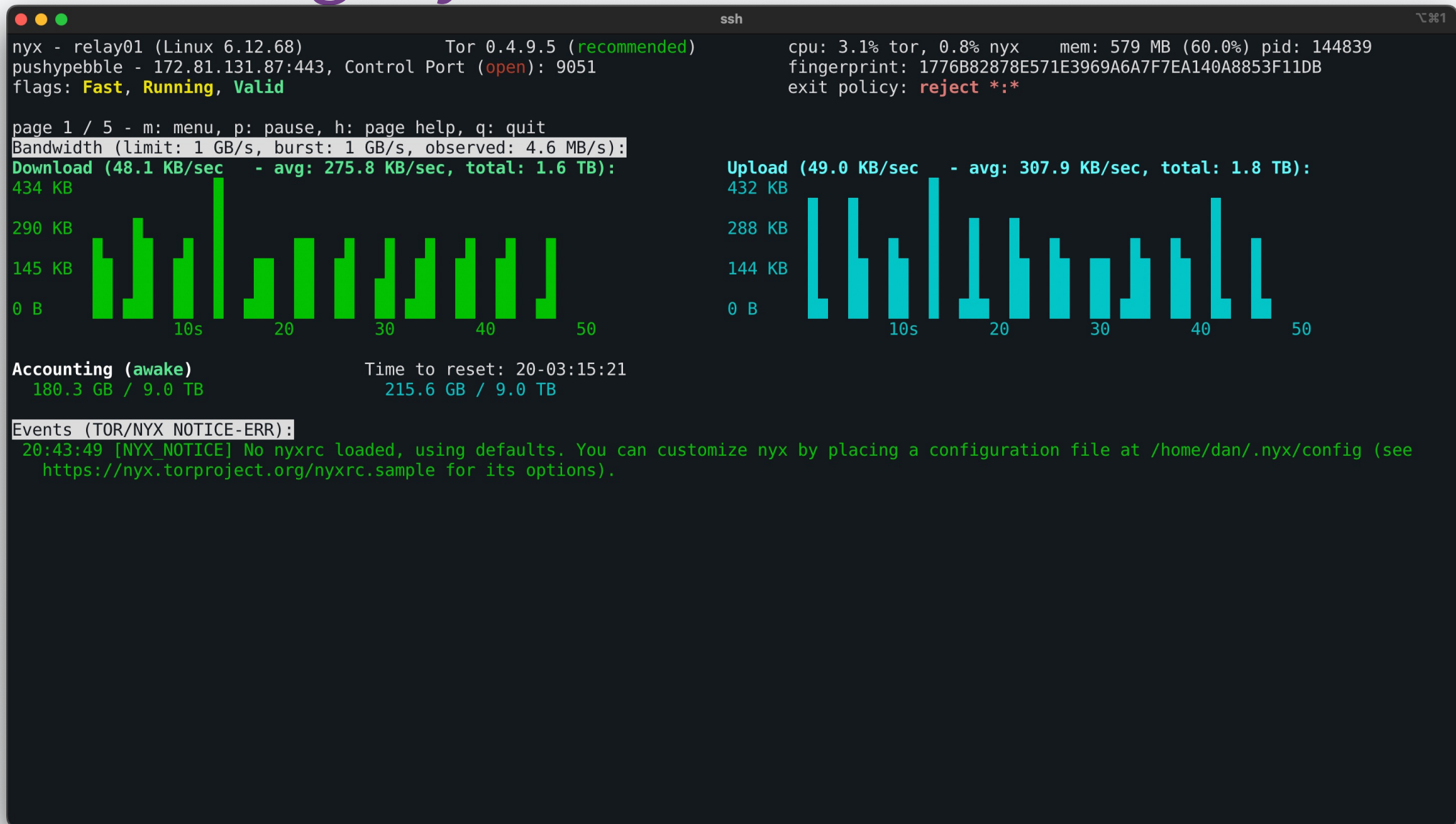
- **Day 68+**

Steady-state guard



<https://blog.torproject.org/lifecycle-of-a-new-relay/>

Monitoring: nyx



Monitoring: nyx

```
ssh
nyx - relay01 (Linux 6.12.68) Tor 0.4.9.5 (recommended) cpu: 1.9% tor, 2.7% nyx mem: 566 MB (58.7%) pid: 144839
pushypebble - 172.81.131.87:443, Control Port (open): 9051 fingerprint: 1776B82878E571E3969A6A7F7EA140A8853F11DB
flags: Fast, Running, Valid exit policy: reject *:.*

page 2 / 5 - m: menu, p: pause, h: page help, q: quit
Connections (1630 outbound, 1 control):
```

172.81.131.87:37864	-->	2.26.16.221:3880 (gb)	A0D680776AB825F9C5C924084130674F155021E4	CoD441	+ 2.9m (OUTBOUND)
172.81.131.87:53958	-->	2.56.10.36:9100 (nl)	7F0541B21603F5D4A17A75C5326CEBC37E8E9D74	DeafCitadel	+ 2.9m (OUTBOUND)
172.81.131.87:36972	-->	2.58.52.163:9003 (de)	8FDAFC439E2D09659A3EC0D723B9C3761EE5596D	SomeTorinAS200462	+ 2.9m (OUTBOUND)
172.81.131.87:57270	-->	2.58.56.43:8430 (nl)	E1EEDB14EFD7AE972D9E4E37C0DB94D1DA630749	Quetzalcoatl	+ 2.9m (OUTBOUND)
172.81.131.87:56194	-->	2.58.56.46:110 (nl)	D20BD0A76403667242FC14732AD1D2596FDC7F46	Quetzalcoatl	+ 2.9m (OUTBOUND)
172.81.131.87:58550	-->	2.58.56.46:8100 (nl)	7220CB589D48AA2872D728A5F3ACC532DB2E4B79	Quetzalcoatl	+ 2.9m (OUTBOUND)
172.81.131.87:57792	-->	2.58.56.92:9400 (nl)	A0C8F635C4644855174CD8C4F09669E532C8A9AB	prsv	+ 2.9m (OUTBOUND)
172.81.131.87:49492	-->	5.9.14.30:143 (de)	0947B12441F69611D7DF7FA94D9188F7AA2B79FF	TorHet3	+ 2.9m (OUTBOUND)
172.81.131.87:45680	-->	5.146.180.12:9001 (de)	195C490D32A3680C275844DF31B12A0C80627078	nodeofmarceldavis	+ 2.9m (OUTBOUND)
172.81.131.87:51406	-->	5.181.80.181:9001 (bg)	79A736CF9E509FD79A39ACC24A3F4AF7AD103533	TheTerrorWorld2	+ 2.9m (OUTBOUND)
172.81.131.87:47224	-->	5.181.156.52:443 (md)	AC2D10FD514ED4CC417BCDD5773A1BCEDAC324D5	pre45	+ 2.9m (OUTBOUND)
172.81.131.87:41518	-->	5.187.242.10:110 (hu)	11298B7CD17A89B34150B95D7E2C63CB8593C307	FOSSonly	+ 2.9m (OUTBOUND)
172.81.131.87:34432	-->	5.199.162.114:9001 (lt)	1331D4F20F9679FEA9F1AAFA2EA9C992AC6F172E	QuakePH0B0S	+ 2.9m (OUTBOUND)
172.81.131.87:46562	-->	5.230.38.235:443 (de)	81E8A4F52F14178E840349FAF92C97AC998DB326	kryptov	+ 2.9m (OUTBOUND)
172.81.131.87:36418	-->	5.250.186.160:9001 (es)	AAE3E83DB1483D40A50E7BE6078F0EDB4D0546BE	magic848relay	+ 2.9m (OUTBOUND)
172.81.131.87:40076	-->	5.252.154.89:9001 (ua)	677D655CBFC52472DAAC850A0A047A646DAAC19A	povostong	+ 2.9m (OUTBOUND)
172.81.131.87:50920	-->	5.254.118.189:9001 (ro)	234B06E4DA915D527C897E21FD1E9F08012E2962	Pizzahut	+ 2.9m (OUTBOUND)
172.81.131.87:56860	-->	5.254.118.191:9001 (ro)	A7AA32CA10E1BEDA1E976A659FD96CA99C2D86FE	Dominos	+ 2.9m (OUTBOUND)
172.81.131.87:36124	-->	5.254.118.192:9001 (ro)	CC2F7932C99F9878A34EEEE01A9089B4EAC36C8FB	PapaJohns	+ 2.9m (OUTBOUND)
172.81.131.87:48000	-->	5.255.111.52:443 (nl)	F163F3E0974A540A32FA1294CDEE7C99F7E04A06	Pia	+ 2.9m (OUTBOUND)
172.81.131.87:54812	-->	5.255.111.104:9001 (nl)	0E20E3BFB1BDC3422E568403E37C00249D9BA901	litserver	+ 2.9m (OUTBOUND)
172.81.131.87:53172	-->	5.255.115.48:9001 (nl)	A22F6515E4ABA5A5B9D280F9296530A2198C6E22	169481492703576	+ 2.9m (OUTBOUND)
172.81.131.87:36396	-->	5.255.117.134:9001 (nl)	5C0433FD8F7C30FB8BF99AB7CA63F3DB43226935	5273156567782171	1.7m (OUTBOUND)
172.81.131.87:42420	-->	5.255.119.76:9100 (nl)	E8303F2E8F6177B8E302A34AE9FE36C19C048492	Camellites	+ 2.9m (OUTBOUND)
172.81.131.87:60342	-->	8.219.48.199:9001 (sg)	CB1F810B21B407B6B1BFB1D7260D220D3F75C07E	Unnamed	+ 2.9m (OUTBOUND)
172.81.131.87:45924	-->	15.204.31.220:443 (us)	D2EBB84C78B98BE128413C1DF4A26652233FC5CD	HDpNGmau7e4WFBKc	+ 2.9m (OUTBOUND)
172.81.131.87:53832	-->	15.204.57.248:443 (us)	F3A7DD1B9C818B90F88C5770C185A84438CC7647	Echidna1	+ 2.9m (OUTBOUND)
172.81.131.87:43254	-->	15.204.220.109:8443 (us)	88BCA446532862CD8C71760BBBAADC5E4DBBE3E2	Art3mis	+ 2.9m (OUTBOUND)
172.81.131.87:33382	-->	23.94.2.177:9954 (us)	12C7D34D22DB0845B24E128E956C14DE676F7A3F	opencommunity	+ 2.9m (OUTBOUND)
172.81.131.87:49108	-->	23.94.152.45:443 (us)	6FD62FC9C9281D99C35BF41BAD9C59E1A0705196	carmine	+ 2.9m (OUTBOUND)
172.81.131.87:33082	-->	23.105.163.117:443 (us)	27132CE932038B4F34F3BB50299C8D8805735512	catpain	+ 2.9m (OUTBOUND)
172.81.131.87:48630	-->	23.108.55.71:443 (us)	DF885E50651903A212EE519BE5A58397AD2E0975	hamburger	+ 2.9m (OUTBOUND)
172.81.131.87:49762	-->	23.111.179.34:626 (us)	D8514224881C60572E06351F2BD650BA14ACA75C	experiment626	+ 2.9m (OUTBOUND)

Monitoring: vnstat

```
dan@relay01:~/ > vnstat
Database updated: 2026-05-11 21:05:00

ens3 since 2023-10-02

      rx:  68.80 TiB      tx:  75.24 TiB      total: 144.04 TiB

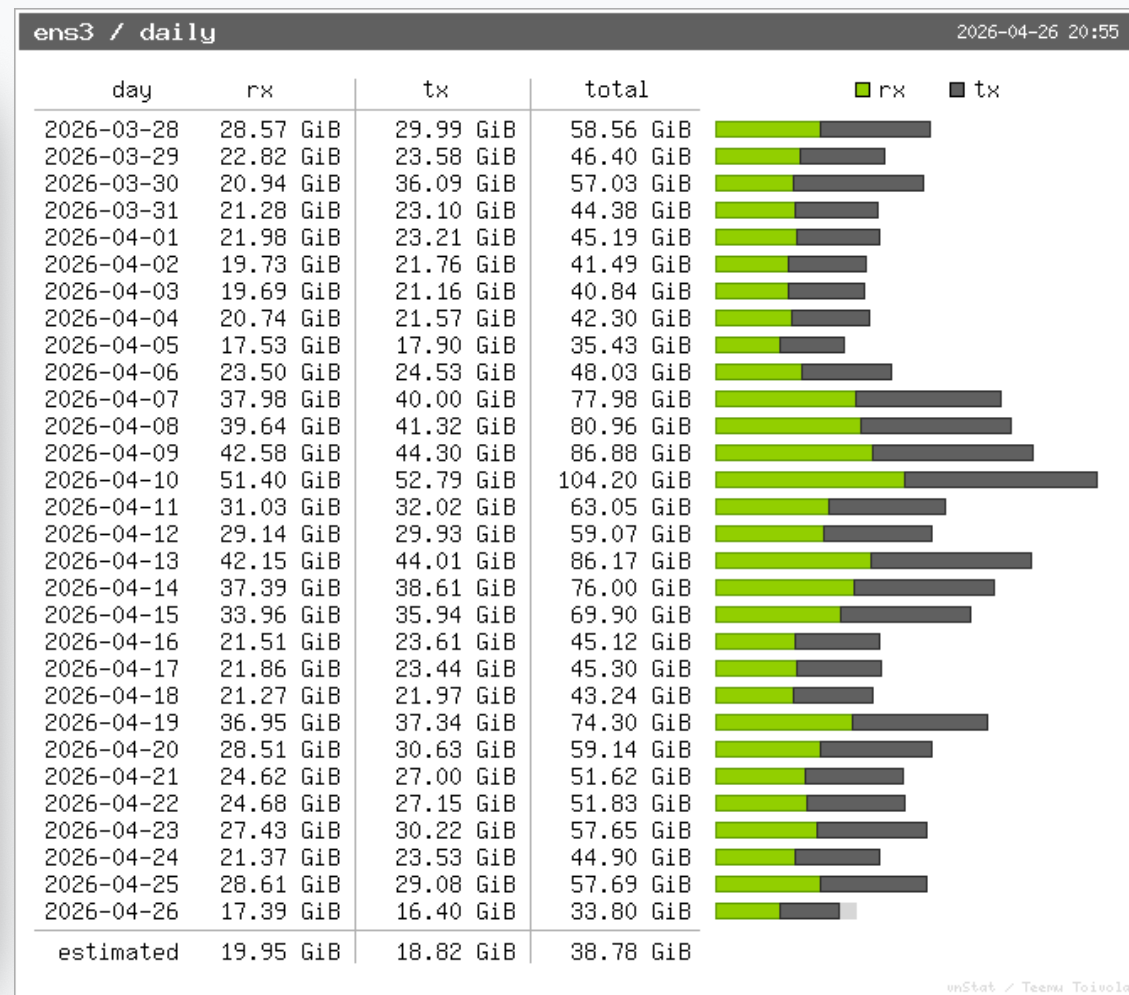
monthly

      rx      |      tx      |      total      |      avg. rate
-----+-----+-----+-----
2026-04      | 827.29 GiB | 879.12 GiB | 1.67 TiB | 5.66 Mbit/s
2026-05      | 212.00 GiB | 231.12 GiB | 443.12 GiB | 4.05 Mbit/s
-----+-----+-----+-----
estimated    | 604.13 GiB | 658.60 GiB | 1.23 TiB |

daily

      rx      |      tx      |      total      |      avg. rate
-----+-----+-----+-----
yesterday    | 13.86 GiB | 13.83 GiB | 27.68 GiB | 2.75 Mbit/s
today        | 23.34 GiB | 25.21 GiB | 48.54 GiB | 5.49 Mbit/s
-----+-----+-----+-----
estimated    | 26.57 GiB | 28.69 GiB | 55.26 GiB |

dan@relay01:~/ > |
```



Post-Install

- Backup Tor Identity Keys in `$DataDirectory/keys`
- Subscribe to `tor-announce` mailing list

Maintenance

- Run regular updates
- Choose how much tinkering you want to do
 - Setup system monitoring & alerting
 - Throughput tuning
- Choose how involved in the community to be
 - Improve Tor project site
 - Discuss feature proposals
 - Run release candidates and report bugs

Help & Community

- [tor-relays](#) mailing list
- [#tor-relays](#) on [irc.oftc.net](#)
- [forum.torproject.org](#)
- Online / In-person operator meetups

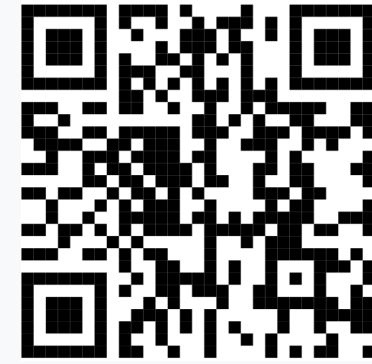
Other Ways to Help

- Tor University Challenge
- Run a snowflake proxy
- Donate to relay associations
 - Shout out Emerald Onion



<https://toruniversity.eff.org>

Thank You



<https://danthesalmon.com/files/2026-tor-talk.pdf>

 dan@infosec.exchange

 [/in/danthesalmon](https://in.danthesalmon)

 [sa7mon](https://github.com/sa7mon)

 danthesalmon.com

“What is Tor” slides adapted from [Understanding How Tor Works](#)
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